Dayoung Kil

Ph.D. student
Department of Intelligent Semiconductor
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I am Dayoung, a Ph.D. student at the Graduate School of Soongsil University, conducting my research under the supervision of Professor Seong Heum Kim in the Visual Intelligence and Platform (VIP) Lab.
My research primarily focuses on hardware-efficient artificial intelligence, with particular emphasis on Efficient Deep Learning, Pruning, Large Vision-Language Models (LVLMs), and Multimodal Large Language Models (MM-LLMs). Recently, I have been actively exploring LVLMs and advanced token pruning techniques to enhance the efficiency and performance of large-scale multimodal systems.

Personal Info

Birth	1998.03.09	GitHub	http://github.com/Dayoung-Kil
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Google Scholar	https://scholar.google.com		

Research Interests

- Computer Vision
- LVLM (Large Vision Language Model)
- MM-LLM (Multi-Modal Large Language Model)
- Pruning
- Efficient Deep Learning
- Vision Transformer

Publications

- 길다영, 김성흠, “단일 파노라마 입력의 실내 공간 레이아웃 복원 모델 경량화,” Journal of Institute of Control, Robotics and Systems(ICROS) 2022 (KCI 등재, SCOPUS)
- 길다영, 김성흠, “Lightweight Room Layout Estimation using a Single Panoramic Image,” International Conference on Control, Automation and Systems(ICCAS) 2022
- Deepak Ghimire, Dayoung Kil and Seong-heum Kim, “A Survey on Efficient Convolutional Neural Networks and Hardware Acceleration,” Electronics 2022 (피인용 횟수 142건 이상)
- Deepak Ghimire, Dayoung Kil and Seong-heum Kim, “A Study of Structure Pruning for Hybrid Neural Networks,” International Conference on Control, Automation and Systems(ICCAS) **2024**
- Deepak Ghimire and Dayoung Kil and Seonghwan Jeong and Jaesik Park and Seong-heum Kim, “One-cycle Structured Pruning with Stability Driven Structure Search”, arXiv 2025

Awards

Feb. 2021	우수상, 숭실 A3+ STAR 코딩 경진대회 AI 모빌리티사업단 of SoongSil University
Feb. 2021	장려상, 제 11회 숭실 캡스톤디자인 경진대회, SoongSil University, Korea “Face Out: A Website that Mosaic People's Faces through face recognition”
Nov. 2021	장려상, AI융합 경진대회, SoongSil University School of AI Convergence, Korea “Face Out: A Website that Mosaic People's Faces through face recognition”
Nov. 2021	우수논문상, 졸업논문발표회, SoongSil University School of AI Convergence, Korea “Lightweight Deep Learning for Room Layout Estimation with a Single Panoramic Image”
Nov. 2024	최우수상(ITTP 원장상), 제 1회 자율주행 인공지능 챌린지, 과학기술정보통신부
May. 2025	4th, 2025 IEEE LOW-POWER COMPUTER VISION CHALLENGE, CVPR Workshop Track1: Image classification for different lighting conditions and styles, Team: VIP

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Patents

- 길다영, 김성흠, “단일 파노라마 입력의 실내 공간 레이아웃 복원 모델 경량화,” C-2022-036602 (소프트웨어등록)

Projects

2021	Face Recognition을 통해 지정된 사람을 제외한 다른 사람들의 얼굴을 모자이크 처리하는 웹 서비스 제작	
2022	Food Detection 및 Volume Estimation을 통해, 이미지 내 음식의 칼로리 계산	
2023	멀티 모달 생성형 AI 기반 가상 의상 및 헤어 피팅 앱 서비스 제작	
2023	시각-언어 모델을 활용한 제로샷 이상 감지 방법론	

Skills

Programming & Tools	Python, C, C++	Anaconda, Jupyter Notebook	Git, VSCode
Deep Learning Frameworks	PyTorch, Torchvision	TensorFlow, Keras	HuggingFace
Computer Vision	OpenCV, Detectron2, Ultralytics YOLO	OpenMMLab (MMDetection, MMSegmentation)	
Web / Platform	HTML, CSS	Android (Java/Kotlin)	

Educations

Mar 2018 – Feb 2022	Bachelor of AI Convergence	Soongil University, Seoul, South Korea
Mar 2022 - Feb 2024	Master of Intelligent Systems	Soongil University, Seoul, South Korea
Mar 2024 – Feb 2027	Doctor of Intelligent Semiconductors	Soongil University, Seoul, South Korea